



Mathematics for Engineering

Exercise Book

CALCULUS

Chapter 1: Function and Limit

1. Find the domain of each function:

a. $f(x) = \sqrt{x+2}$

b. $f(x) = \frac{1}{x^2 - x}$

c. $f(x) = \ln(x+1) - \frac{x}{\sqrt{x-1}}$

2. Find the range of each function:

a. $f(x) = \sqrt{x-1}$

b. $f(x) = x^2 - 2x$

c. $f(x) = \sin x$

3. Determine whether is even, odd, or neither

a. $f(x) = \frac{x}{x^2 + 1}$

b. $f(x) = \frac{x^2}{x^4 + 1}$

c. $f(x) = \frac{x}{x+1}$

4. Explain how the following graphs are obtained from the graph of $f(x)$

a. $f(x-4)$

b. $f(x)+3$

c. $f(x-2)-3$

d. $f(x+5)-4$

5. Suppose that the graph of $f(x) = \sqrt{x}$ is given. Describe how the graph of the function $y = \sqrt{x-1} + 2$ can be obtained from the graph of f .

6. Let $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$. Find each function

a. $f \circ g$

b. $g \circ f$

c. $g \circ g$

d. $f \circ f$

7. Let $f(x) = \frac{x^2 + x + 1}{x}$. Find

a. $f\left(x + \frac{1}{x}\right)$

b. $f(2x-1)$

8. Use the table to evaluate each expression

x	1	2	3	4	5	6
$f(x)$	3	1	4	2	2	5
$g(x)$	6	3	2	1	2	3

a. $f(g(1))$ b. $g(f(1))$ c. $f(f(1))$ d. $g(g(1))$

e. $g \circ f(3)$ f. $g \circ f(6)$

9. Evaluate the following limits

a. $\lim_{x \rightarrow 3} \frac{x^2 + x - 12}{x - 3}$

b. $\lim_{x \rightarrow 1} \frac{x^6 - 1}{x^{10} - 1}$

c. $\lim_{x \rightarrow 0} \frac{\tan 3x}{\tan 5x}$

d. $\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$

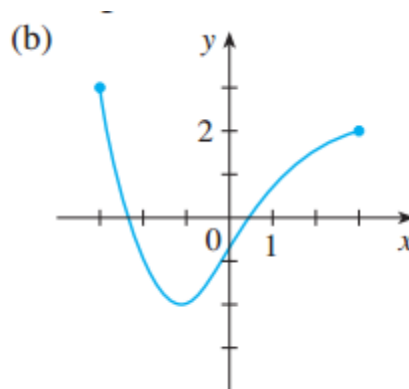
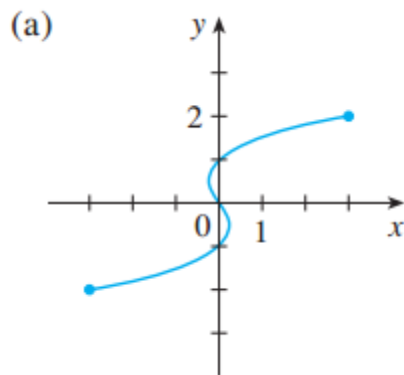
e. $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$

f. $\lim_{x \rightarrow +\infty} \frac{x^2 + x - 12}{x^3 - 3}$

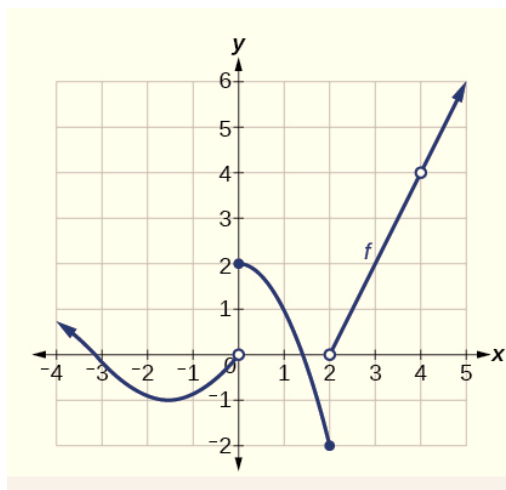
g. $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

h. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{|x - 1|}$

10. Determine whether each curve is the graph of a function of x . If it is, state the domain and range of the function.



11. The graph of f is given.



a. Find each limit, or explain why it does not exist.

i. $\lim_{x \rightarrow 0^+} f(x)$, $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0} f(x)$

iii. $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow 4} f(x)$

b. At what numbers is discontinuous?

12. Determine where the function $f(x)$ is continuous

a. $f(x) = \frac{2x^2 + x - 1}{x - 2}$ b. $f(x) = \frac{x - 9}{\sqrt{4x^2 + 4x + 1}}$ c. $f(x) = \ln(2x + 5)$

13. Find the constant m that makes f continuous on $\mathbb{R}$

a. $f(x) = \begin{cases} x^2 - m^2, & x < 4 \\ mx + 20, & x \geq 4 \end{cases}$

b. $f(x) = \begin{cases} mx^2 + 2x, & x < 2 \\ x^3 - mx, & x \geq 2 \end{cases}$

c. $f(x) = \begin{cases} \frac{e^{2x} - 1}{x}, & x \neq 0 \\ m, & x = 0 \end{cases}$

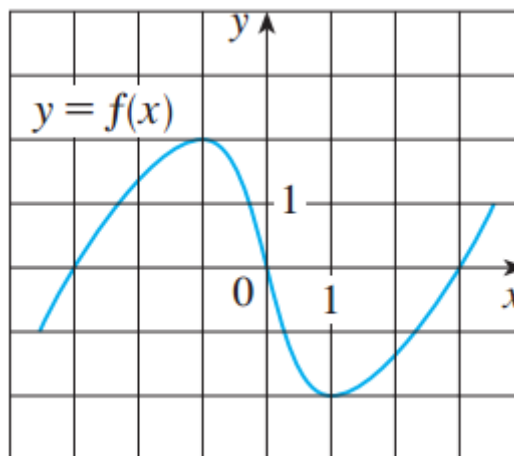
d. $f(x) = \begin{cases} \frac{x^2 - 1}{\sqrt{x} - 1}, & x > 1 \\ mx + 1, & x \leq 1 \end{cases}$

14. Find the numbers at which the function $f(x) = \begin{cases} x + 2, & x < 0 \\ 2x^2, & 1 \geq x \geq 0 \\ 2 - x, & x > 1 \end{cases}$ is discontinuous.

Chapter 2: Derivatives

1. Use the given graph to estimate the value of each derivative

- a. $f'(-3)$ b. $f'(-1)$
c. $f'(0)$ d. $f'(3)$



2. Find an equation of the tangent line to the curve at the given point:

- a. $y = \frac{x-1}{x-2}$, $(3, 2)$ b. $y = \frac{2x}{x^2+1}$, $(0, 0)$
c. $y = 3 - 2x + x^2$, $x = 1$ d. $y = \frac{3-2x}{x-1}$, $y = -1$

3. Find y'

- a. $y = x^2 - x\sqrt{x} + \frac{1}{x} + 2$ b. $y = \sqrt{x + \sqrt{x}}$ c. $y = \frac{x^2}{x+1}$
d. $y = x\sqrt{x+2}$ e. $y = \ln(x^2 + 1) - \frac{1}{x}$ f. $y = e^x \sin(2x+1)$

4. Find y''

- a. $y = xe^{3x-1}$ b. $y = \sqrt[3]{2x+1}$ c. $y = e^{-x} \cos x$

5. Find dy/dt for:

- a. $y = x^3 + x + 2$, $dx/dt = 2$ and $x = 1$ b. $y = \ln x$, $dx/dt = 1$ and $x = e^2$
c. $y = \tan \sqrt{t}$ and $t = \frac{\pi^2}{16}$ d. $\begin{cases} y = \sin \varphi \\ t = \cos \varphi \end{cases}$ and $\varphi = \frac{\pi}{3}$

6. Find dy for:

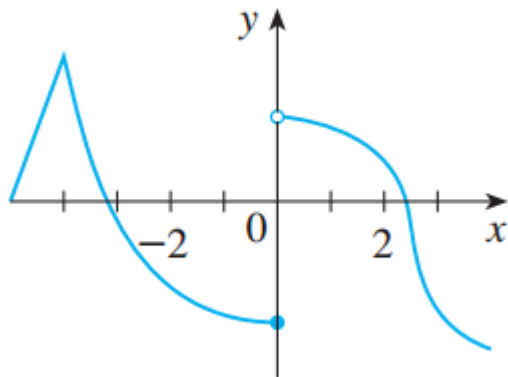
a. $y = \frac{1}{x^2 + 1}$

b. $y = \sqrt{x+1}, x=3$

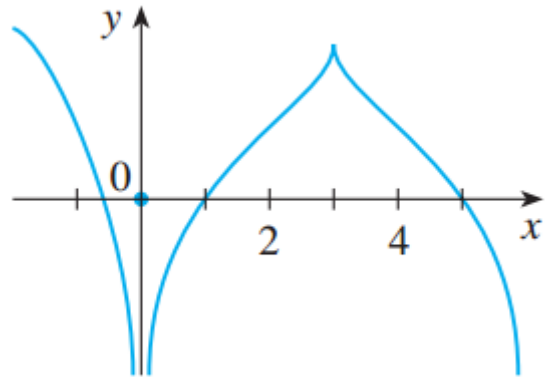
c. $y = \ln(x^2 + 1)$, $x=1$ and $dx=2$

7. The graph of is given. State the numbers at which is not differentiable

a.



b.



8. A table of values for f , f' , g and g' is given

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	4	6
2	1	8	5	7
3	7	2	7	9

a. If $h(x) = f(g(x))$, find $h'(1)$

b. If $H(x) = g \circ f(x)$, find $H'(1)$

c. If $F(x) = f \circ f(x)$, find $F'(2)$

d. If $G(x) = g \circ g(x)$, find $G'(3)$

9. If $h(x) = \sqrt{4 + 3f(x)}$, where $f(1) = 7$, $f'(1) = 4$, find $h'(1)$.

10. For the circle: $x^2 + y^2 = 25$.

a. Find dy/dx

b. Find an equation of the tangent to the circle at the point $(3, 4)$.

11. Let $(L): x^3 + y^3 = 6xy$

a. Find dy/dx

b. Find an equation of tangent to the curve (L) at the point (3, 3)

12. Find y' by implicit differentiation

a. $x^4 + y^4 = 16x + y$ b. $\sqrt{x} + \sqrt{y} = 4$ c. $x^3 + xy = y^2$

13. Find f' in terms of g'

a. $f(x) = g(\sin 2x)$ b. $f(x) = g(e^{1-3x})$

14. Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm^2 ?

15. If $x^2 + y^2 = 25$ and $dy/dt = 6$, find dx/dt when $y = 4$ and $x > 0$.

16. If $z^2 = x^2 + y^2$ ($z > 0$), $dx/dt = 2$, $dy/dt = 3$, find dz/dt when $x = 5$, $y = 12$

17. Find the linearization $L(x)$ of the function at a .

a. $f(x) = \frac{1}{\sqrt{2+x}}$, $a = 2$ b. $f(x) = \sqrt[3]{5-x}$, $a = -3$

18. The equation of motion is $s(t) = 3\sin t - 4\cos t + 1$ for a particle, where s is in meters and t is in seconds. Find the acceleration (in m/s^2) after 3 seconds.

Chapter 3: Applications of Differentiation

1. Find the absolute maximum and absolute minimum values of the function on the given interval

a. $f(x) = 3x^2 - 12x + 5, [0; 3]$

b. $f(x) = x^3 - 3x + 5, [0; 3]$

c. $f(x) = x\sqrt{4 - x^2}, [-1; 2]$

d. $f(x) = x - \ln x, \left[\frac{1}{2}; 2\right]$

2. Find the critical numbers of the function

a. $f(x) = 5x^2 + 4x$

b. $f(x) = \frac{x-1}{x^2 - x + 1}$

c. $f(x) = x \ln x$

3. Find all numbers that satisfy the conclusion of the Rolle's Theorem

a. $f(x) = x\sqrt{x+2}, [-2; 0]$

b. $f(x) = (x-2)x^2, [0; 2]$

4. Find all numbers that satisfy the conclusion of the Mean Value Theorem

a. $f(x) = 3x^2 + 2x + 5, [-1; 1]$

b. $f(x) = e^{-2x}, [0; 3]$

5. If $f(1) = 10$ and $f'(x) \geq 2, \forall x \in [1; 4]$, how small can $f(4)$ possibly be?

6. Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$ is increasing and where it is decreasing.

7. Find the inflection points for the function

a. $f(x) = x^4 - 4x + 1$

b. $f(x) = x^6$

c. $f(x) = xe^x$

8. Find $f(x)$ for $f'(x) = \sqrt{2x+1}$ and $f(0) = 1$.

9. Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1; 4)$

10. Find two numbers whose difference is 100 and whose product is a minimum.

11. Find two positive numbers whose product is 100 and whose sum is a minimum.

12. Use Newton's method with the specified initial approximation x_1 to find x_3

a. $x^3 + 2x - 4 = 0, x_1 = 1$

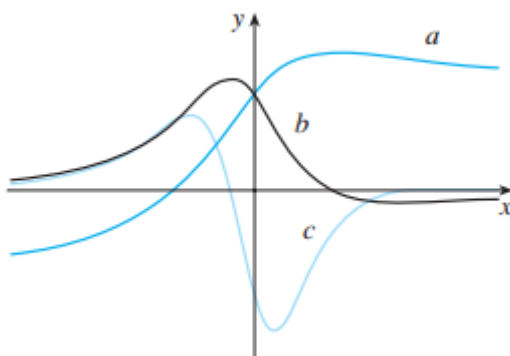
b. $x^5 + 2 = 0, x_1 = -1$

c. $\ln(x^2 + 1) - 2x - 1 = 0, x_1 = 1$

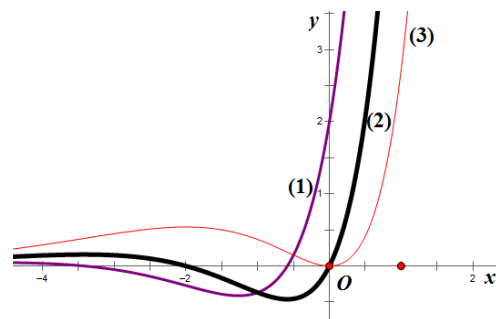
d. $\ln(4 - x^2) = x, x_1 = 1$

13. The figure shows the graphs of f, f' and f'' . Identify each curve, and explain your choices

a.



b.



14. Find the most general anti-derivative of the function.

a. $f(x) = 6x^2 - 2x + 3$

b. $f(x) = \sqrt[6]{x} + \frac{1}{x^2}$

c. $f(x) = \frac{x^2 + x + 2}{x}$

d. $f(x) = 2x(x^2 + 1)$

15. Find the anti-derivative of that satisfies the given condition

a. $f(x) = 5x^4 - 2x^5, F(0) = 4$

b. $f(x) = 4 - \frac{2x}{x^2 + 1}, F(0) = 1$

16. A particle is moving with the given data. Find the position of the particle

a. $v(t) = \sin t - \cos t, s(0) = 0$

b. $v(t) = 10\sin t + 3\cos t, s(\pi) = 0$

c. $v(t) = 10 + 3t - 3t^2, s(2) = 10$

Chapter 4 - 6: Integration

1. Estimate the area under the graph of $y = f(x)$ using 6 rectangles and left endpoints

a. $f(x) = \frac{1}{x} + x$, $x \in [1, 4]$

b. $f(x) = x^2 - 2$, $x \in [-1, 2]$

c. A table of values for f is given

x	1	2	3	4	5	6	7
$f(x)$	5	6	3	2	7	1	2

3. Repeat part (1) using right endpoints

4. For the function $f(x) = x^3$, $x \in [-2, 2]$. Estimate the area under the graph of using four approximating rectangles and taking the sample points to be

a. Right endpoints

b. Left endpoints

c. Midpoints

5. Use (a) the Trapezoidal Rule, (b) the Midpoint Rule, and (c) Simpson's Rule to approximate the given integral with the specified value of n .

a. $\int_0^3 \sqrt{x} dx$, $n = 4$

b. $\int_1^3 \frac{\sin x}{x} dx$, $n = 6$

6. Let $I = \int_0^2 \frac{dx}{x^2 + 1}$. Find the approximations L_4 , R_4 , M_4 , T_4 and S_4 for I .

7. Find the derivative of the function $g(x) = \int_0^x \sqrt{t^2 + 1} dt$

8. Find g'

a. $g(x) = \int_1^{x^4} \frac{1}{\cos t} dt$

b. $g(x) = \int_1^{\sqrt{x}} \frac{\sin u}{u} du$

$$c. g(x) = \int_{2x}^{x^2+x+2} \frac{e^t}{t} dt$$

$$d. g(x) = \int_{\sin x}^{\cos x} (1+v^2)^{10} dv$$

9. Find the average value of the function on the given interval

$$a. f(x) = x^2, \quad [-1, 1]$$

$$b. f(x) = \frac{1}{x}, \quad [1, 5]$$

$$c. f(x) = x\sqrt{x}, \quad [1, 4]$$

$$d. f(x) = x \ln x, \quad [1, e^2]$$

10. A particle moves along a line so that its velocity at time t is $v(t) = t^2 - t - 6$ (m/s)

a. Find the displacement of the particle during the time period $1 \leq t \leq 4$

b. Find the distance traveled during this time period

11. Suppose the acceleration function and initial velocity are $a(t) = t + 3$ (m/s²), $v(0) = 5$ (m/s). Find the velocity at time t and the distance traveled when $0 \leq t \leq 5$.

12. A particle moves along a line with velocity function $v(t) = t^2 - t$, where t is measured in meters per second. Find the displacement and the distance traveled by the particle during the time interval $t \in [0, 2]$.

13. Evaluate the integral

$$a. \int_0^2 x^2 \cdot \sqrt{x^3 + 1} dx$$

$$b. \int x e^{x^2} dx$$

$$c. \int \left(\frac{1}{x} + \sqrt{x} - 3x^2 \right) dx$$

$$d. \int_0^1 y(1+y^2)^5 dy$$

$$e. \int \frac{\ln x}{x} dx$$

$$f. \int \frac{t}{t^2 + 1} dt$$

14. Evaluate the integral

$$a. \int x e^x dx$$

$$b. \int_0^1 x^2 e^{-x} dx$$

$$c. \int x \sin x dx$$

$$d. \int \ln x dx$$

$$e. \int_1^e x \ln x dx$$

$$f. \int e^{\sqrt{x}} dx$$

15. Suppose $f(x)$ is differentiable, $f(1) = 4$ and $\int_0^1 f(x) dx = 5$. Find $\int_0^1 xf'(x) dx$

16. Suppose $f(x)$ is differentiable, $f(1) = 3$, $f(3) = 1$ and $\int_1^3 xf'(x) dx = 13$. What is the average value of f on the interval $[1,3]$?

17. Let $f(x) = \begin{cases} -x-1, & -3 \leq x \leq 0 \\ -\sqrt{1-x^2}, & 0 < x \leq 1 \end{cases}$. Evaluate $\int_{-3}^1 f(x) dx$

18. Find $g'(0)$ for

a. $g(x) = \int_x^{x^2} e^{2t+1} dt$ b. $\int_{2x-1}^{x^3} u\sqrt{u+1} du$

19. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a. $\int_1^{\infty} \frac{dx}{(3x+1)^2}$ b. $\int_{-\infty}^0 \frac{dx}{2x-5}$ c. $\int_{-\infty}^{-1} \frac{dx}{\sqrt{2-x}}$ d. $\int_0^{\infty} \frac{xdx}{(x^2+2)^2}$

e. $\int_4^{\infty} e^{\frac{y}{2}} dy$ f. $\int_{-\infty}^{-1} e^{-2t} dt$ g. $\int_{2\pi}^{\infty} \sin \phi d\phi$ h. $\int_{-\infty}^{\infty} xe^{-x^2} dx$

i. $\int_0^1 \frac{dx}{4x-1}$ j. $\int_3^4 \frac{dx}{\sqrt{x-3}}$ k. $\int_{-1}^1 \frac{1}{\sqrt[3]{x^2}} dx$ l. $\int_0^1 \frac{dx}{\sqrt{x}}$

20. Use the Comparison Theorem to determine whether the integral is convergent or divergent

a. $\int_1^{\infty} \frac{\cos^2 x dx}{1+x^2}$ b. $\int_1^{\infty} \frac{2+e^{-x}}{x} dx$ c. $\int_1^{\infty} \frac{dx}{x+e^{2x}}$

d. $\int_1^{\infty} \frac{xdx}{\sqrt{1+x^6}}$ e. $\int_0^{\frac{\pi}{2}} \frac{\cos x dx}{\sqrt{\sin x}}$ f. $\int_0^1 \frac{2dx}{\sqrt{x^3}}$

LINEAR ALGEBRA

Chapter 1: Systems of Linear Equations

1. Write the augmented matrix for each of the following systems of linear equations and then solve them.

$$\text{a. } \begin{cases} -x + y + 2z = 1 \\ 2x + 3y + z = -2 \\ 5x + 4y + 2z = 4 \end{cases}$$

$$\text{b. } \begin{cases} 2x + 3y + z = 10 \\ 2x - 3y - 3z = 22 \\ 4x - 2y + 3z = -2 \end{cases}$$

$$\text{c. } \begin{cases} x + y + z = 0 \\ 2x - y + 2z = 0 \\ x + z = 0 \end{cases}$$

$$\text{d. } \begin{cases} x_1 + 2x_2 - x_3 + x_4 = 0 \\ 2x_1 + 3x_2 - 2x_3 + 3x_4 = 0 \\ x_1 + x_2 - 3x_3 + x_4 = 0 \end{cases}$$

2. Compute the rank of each of the following matrices.

$$\text{a. } A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{pmatrix}$$

$$\text{b. } B = \begin{pmatrix} -2 & 3 & 3 \\ 3 & -4 & 1 \\ -5 & 7 & 2 \end{pmatrix}$$

$$\text{c. } C = \begin{pmatrix} 1 & 1 & -1 & 4 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & 5 & 8 \end{pmatrix}$$

$$\text{d. } D = \begin{pmatrix} 1 & 1 & -1 & 3 \\ -1 & 4 & 5 & -2 \\ 1 & 6 & 3 & 4 \end{pmatrix}$$

3. Find all values of k for which the system has nontrivial solutions and determine all solutions in each case.

$$\text{a. } \begin{cases} x - y + 2z = 0 \\ -x + y - z = 0 \\ x + ky + z = 0 \end{cases}$$

$$\text{b. } \begin{cases} x - 2y + z = 0 \\ x + ky - 3z = 0 \\ x - 6y + 5z = 0 \end{cases}$$

$$\text{c. } \begin{cases} x + y + z = 0 \\ x + y - z = 0 \\ x + y + kz = 0 \end{cases}$$

$$\text{d. } \begin{cases} x + y - z = 0 \\ ky - z = 0 \\ x + y + kz = 0 \end{cases}$$

4. Determine the values of m such that the system of linear equations has exactly one solution.

$$\text{a. } \begin{cases} x - y + 2z = m \\ -x + y - z = 0 \\ -x + my - z = 1 - m \end{cases}$$

$$\text{b. } \begin{cases} mx + y + z = 1 \\ x + my + z = m \\ x + y + mz = m^2 \end{cases}$$

$$\text{c. } \begin{cases} x + y - z = 1 \\ x + my + 2z = m \\ x + 2y + z = 2 \end{cases}$$

$$\text{d. } \begin{cases} x + my - mz = m \\ 2x + y - z = 2 \\ x + y + z = 0 \end{cases}$$

5. Determine the values of m such that the system of linear equations is inconsistent.

$$\text{a. } \begin{cases} x - y + 2z = m \\ -x + y - z = 0 \\ x - y + 3z = 1 - m \end{cases}$$

$$\text{b. } \begin{cases} x - 2y + 2z = m \\ x + my - z = 0 \\ 2x + y + mz = 2 - m \end{cases}$$

6. Find a , b and c so that the system $\begin{cases} x + ay + cz = 0 \\ bx + cy - 3z = 1 \\ ax + 2y + bz = 5 \end{cases}$ has the solution $(3, -1, 2)$

7. Consider the matrix $A = \begin{pmatrix} 2 & -1 & 3 \\ -4 & 2 & k \\ 4 & -2 & 6 \end{pmatrix}$

a. If A is the augmented matrix of a system of linear equations, determine the number of equations and the number of variables.

b. If A is the augmented matrix of a system of linear equations, find the value(s) of k such that the system is consistent.

8. Find all values of k so that the system of equations has no solution.

$$\text{a. } \begin{cases} x + y - z = 2 \\ -2y + z = 3 \\ 4y - 2z = k \end{cases}$$

$$\text{b. } \begin{cases} x + y - z = 1 \\ 2x + (k + 5)y - 2z = 4 \\ x + (k + 3)y + (k - 1)z = k + 3 \end{cases}$$

9. Find all values of a and b for which the system of equations
$$\begin{cases} x + y + 3z = 2 \\ x + 2y + 5z = 1 \\ 2x + 2y + az = b \end{cases}$$
 is inconsistent.

10. Solve the system of linear equation corresponding to the given augmented matrix

a. $A = \begin{pmatrix} 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$ b. $B = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

11. Determine the values of m such that the rank of the matrix is 2

A. $\begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 5 \\ 1 & 2 & m \end{pmatrix}$ b. $\begin{pmatrix} 1 & 2 & 1 & 4 \\ 2 & 1 & 1 & 5 \\ -3 & 6 & 1 & m \end{pmatrix}$ c. $\begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & -1 \\ 3 & 1 & 2 \\ m & 3 & 5 \end{pmatrix}$

12. Solve the system
$$\begin{cases} x + 2y = 12 \\ 3x - y = 8 \\ -x + 5y = 16 \end{cases}$$

Chapter 2: Matrix Algebra

1. Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 1 \\ 3 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 3 & -4 \\ -1 & 2 & 1 \end{pmatrix}$. Compute the matrix

- a. $2A - B^T$ b. AB c. BA d. AC
 e. CC^T f. C^TC g. A^3 h. B^2A^T

2. Suppose that A and B are nxn matrices. Simplify the expression

- a. $(A+B)^2 - (A-B)^2$ b. $A(BC - CD) + A(C - B)D - AB(C - D)$

3. Let $A = \begin{pmatrix} 3 & 1 & 2 \\ 4 & 8 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 5 & 2 & 1 \\ 1 & 8 & 0 & -6 \\ 1 & 4 & 3 & 7 \end{pmatrix}$.

a. Compute AB

b. Compute $f(A)$ if $f(x) = x^2 - 3x + 2$

4. Find the inverse of each of the following matrices.

- a. $\begin{pmatrix} 1 & 5 \\ 2 & -1 \end{pmatrix}$ b. $\begin{pmatrix} 2 & 1 \\ 2 & -4 \end{pmatrix}$ c. $\begin{pmatrix} 1 & -1 & 2 \\ -5 & 7 & -11 \\ -2 & 3 & -5 \end{pmatrix}$ d. $\begin{pmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{pmatrix}$

5. Given $A^{-1} = \begin{pmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{pmatrix}$. Find a matrix X such that

- a. $AX = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ b. $AX = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ c. $XA = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 1 & 1 \end{pmatrix}$

6. Find A when

$$\text{a. } (3A)^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix} \quad \text{b. } (I + 2A)^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \quad \text{c. } (A^{-1} - 2I)^T = -2 \begin{pmatrix} 1 & 4 \\ 3 & 11 \end{pmatrix}$$

7. Write the system of linear equations in matrix form and then solve them.

$$\text{a. } \begin{cases} 2x - y = 4 \\ 3x + 2y = -4 \end{cases} \quad \text{b. } \begin{cases} 2x + 3y + z = 10 \\ 2x - 3y - 3z = 22 \\ 4x - 2y + 3z = -2 \end{cases} \quad \text{c. } \begin{cases} x + y = a \\ 2x + 3y = 1 - 2a \end{cases} (a \in \mathbb{R})$$

8. Find A^{-1} if

$$\text{a. } A^2 - 6A + 5I = 0 \quad \text{b. } A^2 + 3A - I = 0 \quad \text{c. } A^4 = I$$

9. Solve for X

$$\text{a. } \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} X = \begin{pmatrix} 1 & -1 \\ 3 & 3 \end{pmatrix} \quad \text{b. } ABXC = B^T \quad \text{c. } AX^T BC = B$$

(where A , B and C are $n \times n$ invertible matrices)

$$10. \text{ Compute } \begin{pmatrix} -1 & 3 \\ 0 & 1 \end{pmatrix}^{101}$$

11. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, and assume that $T(1, 2) = (-1, 1)$ and $T(0, 3) = (-3, 3)$

$$\text{a. Compute } T(11, -5) \quad \text{b. Compute } T(1, 11)$$

$$\text{c. Find the matrix of } T \quad \text{d. Compute } T^{-1}(2, 3)$$

12. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that the matrix of T is $\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$.

Find $T(3, -2)$

$$13. \text{ The } (2; 1)\text{-entry of the product } \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 5 & 1 \\ 4 & -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 2 \\ 5 & 1 & 0 \\ 0 & 4 & 3 \end{pmatrix}$$

Chapter 3: Determinants and Diagonalization

1. Evaluate the determinant

$$\text{a. } \begin{vmatrix} x-2 & -1 \\ -3 & x \end{vmatrix} \quad \text{b. } \begin{vmatrix} -2 & 0 & 0 \\ 4 & 6 & 0 \\ -3 & 7 & 2 \end{vmatrix} \quad \text{c. } \begin{vmatrix} -3 & 2 & 1 \\ 4 & 5 & 6 \\ 2 & -3 & 1 \end{vmatrix} \quad \text{d. } \begin{vmatrix} 2 & -1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 4 \end{vmatrix}$$

$$\text{e. } \begin{vmatrix} x & y & 1 \\ -1 & -2 & 1 \\ 1 & 5 & 1 \end{vmatrix} \quad \text{f. } \begin{vmatrix} m & -1 & 0 \\ 1 & 2 & 1 \\ 2 & m & -3 \end{vmatrix}$$

2. Find the minors and the cofactors of the matrix

$$\text{a. } A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} \quad \text{b. } B = \begin{pmatrix} -3 & 4 & 2 \\ 6 & 3 & 1 \\ 4 & -7 & -8 \end{pmatrix} \quad \text{c. } C = \begin{pmatrix} 1 & 3 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & m \end{pmatrix}$$

3. Find the adjugate and the inverse of the matrix $A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{pmatrix}$

$$\text{4. Let } A = \begin{pmatrix} 1 & * & * & * \\ 0 & -1 & * & * \\ 0 & 0 & 2 & * \\ 0 & 0 & 0 & 2 \end{pmatrix}. \text{ Find}$$

$$\begin{array}{lll} \text{a. } |2A^{-1}| & \text{b. } |AA^T| & \text{c. } |\text{adj } A| \\ \text{d. } |-A^3| & \text{e. } |(2A)^{-1}| & \text{f. } |A^{-1} - 2\text{adj } A| \end{array}$$

5. Let A and B be square matrices of order 4 such that $|A| = -5$ and $|B| = 3$. Find

$$\text{a. } |2AB| \quad \text{b. } |\text{adj}(AB)| \quad \text{c. } |5A^{-1}B^T| \quad \text{d. } |A^T B^{-1} A^2|$$

6. Find all values of k for which the matrix is not invertible

$$\text{a. } A = \begin{pmatrix} 1 & 3 \\ k & 2 \end{pmatrix} \quad \text{b. } B = \begin{pmatrix} m & 1 & 3 \\ 1 & 3 & 2 \\ -1 & 4 & 5 \end{pmatrix} \quad \text{c. } C = \begin{pmatrix} m & 2 & 0 \\ 1 & m & 1 \\ 2 & 3 & 1 \end{pmatrix}$$

7. Find the **characteristic polynomial** of the matrix

$$\text{a. } A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \quad \text{b. } B = \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix}$$

$$\text{c. } C = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix} \quad \text{d. } D = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ -1 & 1 & 1 \end{pmatrix}$$

8. Find the eigenvalues and corresponding eigenvectors of the matrix

$$\text{a. } A = \begin{pmatrix} -3 & 5 \\ 10 & 2 \end{pmatrix} \quad \text{b. } B = \begin{pmatrix} 5 & 4 \\ 2 & 1 \end{pmatrix}$$

$$\text{c. } C = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \quad \text{d. } D = \begin{pmatrix} -3 & 2 & -1 \\ 0 & 1 & 0 \\ 4 & 1 & 1 \end{pmatrix}$$

9. Find the determinant of the matrix $A = \begin{pmatrix} 5 & 1 & 2 & 4 \\ 1 & 0 & -1 & -3 \\ 1 & 1 & 6 & 1 \\ 1 & 0 & 0 & -4 \end{pmatrix}$

10. Find the (1, 2)-cofactor and (3,1) - cofactor of the matrix $\begin{bmatrix} -1 & 3 & -2 \\ 4 & 5 & -7 \\ 7 & 8 & 1 \end{bmatrix}$

11. Let $A = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 0 \\ 2 & -1 & x \end{pmatrix}$. For which values of x is A invertible ?

Chapter 4: Vector Geometry

1. Find the equations of the line through the points $P_0(2, 0, 1)$ and $P_1(4, -1, 1)$.
2. Find the equations of the line through $P_0(3, -1, 2)$ parallel to the line with equations:

$$\begin{cases} x = -1 + 2t \\ y = 1 + t \\ z = -3 + 4t \end{cases}$$

3. Determine whether the following lines intersect and, if so, find the point of intersection.

$$\begin{cases} x = 1 - 3t \\ y = 2 + 5t \\ z = 1 + t \end{cases}, \begin{cases} x = -1 + s \\ y = 3 - 4s \\ z = 1 - s \end{cases}$$

4. Compute $\|v\|$ if v equals:

a. $(2, -1, 2)$ b. $2(1, 1, -1)$ c. $-3(1, 1, 2)$ d. $(1, 2, 3) - (4, 1, 2)$

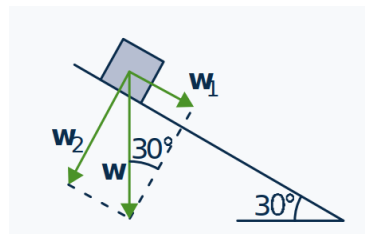
5. Find a unit vector in the direction from $(3, -1, 4)$ to $(1, 3, 5)$.

6. Find $\|v - 3w\|$ when $\|v\| = 2$, $\|w\| = 1$, and $v \cdot w = 2$

7. Compute the angle between $u = (-1, 1, 2)$ and $v = (-1, 2, 1)$.

8. Show that the points $P(3, -1, 1)$, $Q(4, 1, 4)$, and $R(6, 0, 4)$ are the vertices of a right triangle.

9. Suppose a ten-kilogram block is placed on a flat surface inclined 30° to the horizontal as in the diagram. Neglecting friction, how much force is required to keep the block from sliding down the surface?



10. Find the projection of $u = (2, -3, 1)$ on $d = (-1, 1, 3)$ and express $u = u_1 + u_2$ where u_1 is parallel to d and u_2 is orthogonal to d .

11. Find an equation of the plane through $P_0(1, -1, 3)$ with $n = (-3, -1, 2)$ as normal.

12. Find an equation of the plane through $P_0(3, -1, 2)$ that is parallel to the plane with equation $2x - 3y - z = 6$.

13. Find the shortest distance from the point $P(2, -1, -3)$ to the plane with equation $3x - y + 4z = 1$. Also find the point Q on this plane closest to P .

14. Find the equation of the plane through $P(1, 3, -2)$, $Q(1, 1, 5)$, and $R(2, -2, 3)$.

15. Find the shortest distance between the nonparallel lines

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

16. Compute $u \cdot v$ where:

a. $u = (2, -1, 3)$, $v = (-1, 1, 1)$ b. $u = (-2, 1, 4)$, $v = (-1, 5, 1)$

17. Find all real numbers x such that:

a. $(3, -1, 2)$ and $(3, -2, x)$ are orthogonal.

b. $(2, -1, 1)$ and $(1, x, 2)$ are at an angle of $\pi/3$.

18. Find the three internal angles of the triangle with vertices:

a. $A(3, 1, -2)$, $B(3, 0, -1)$, and $C(5, 2, -1)$

b. $A(3, 1, -2)$, $B(5, 2, -1)$, and $C(4, 3, -3)$

19. Find the equations of the line of intersection of the following planes.

a. $2x - 3y + 2z = 5$ and $x + 2y - z = 4$.

b. $3x + y - 2z = 1$ and $x + y + z = 5$.

20. Find the area of the triangle with vertices $P(2, 1, 0)$, $Q(3, -1, 1)$, and $R(1, 0, 1)$

21. Find the volume of the parallelepiped determined by the vectors $u = (1, 2, -1)$, $v = (3, 4, 5)$ and $w = (-1, 2, 4)$.

22. In each case show that that T is either projection on a line, reflection in a line, or rotation through an angle, and find the line or angle

a. $T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} x+2y \\ 2x+4y \end{bmatrix}$

b. $T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{2} \begin{bmatrix} x-y \\ y-x \end{bmatrix}$

c. $T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -x-y \\ x-y \end{bmatrix}$

d. $T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -3x+4y \\ 4x+3y \end{bmatrix}$

e. $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ -x \end{bmatrix}$

f. $T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{2} \begin{bmatrix} x-\sqrt{3}y \\ \sqrt{3}x+y \end{bmatrix}$

23. Determine the effect of the following transformations.

a. Rotation through $\pi/2$, followed by projection on the y axis, followed by reflection in the line $y = x$.

b. Projection on the line $y = x$ followed by projection on the line $y = -x$.

c. Projection on the x axis followed by reflection in the line $y = x$.

24. Find the reflection of the point P in the line $y = 1 + 2x$ in $\mathbb{R}^2$ if:

a. $P = P(1, 1)$

b. $P = P(1, 4)$

25. Find the angle between the following pairs of vectors.

a. $u = (1, -1, 4)$, $v = (5, 2, -1)$

b. $u = (2, 1, 5)$, $v = (0, 3, 1)$

26. In each case, compute the projection of $\mathbf{u}$ on $\mathbf{v}$.

a. $\mathbf{u} = \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$

b. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$

c. $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

d. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -6 \\ 4 \\ 2 \end{bmatrix}$

27. Find the shortest distance between the following pairs of nonparallel lines and find the points on the lines that are closest together.

a. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix};$
 $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

b. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix};$
 $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$

c. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix};$
 $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$

d. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + s \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix};$
 $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Chapter 5: The Vector Space $\mathbb{R}^n$

1. Let $x = (-1, -2, -2)$, $u = (0, 1, 4)$, $v = (-1, 1, 2)$ and $w = (3, 1, 2)$ in $\mathbb{R}^3$. Find scalars a , b and c such that $x = au + bv + cw$

2. Write v as a linear combination of u and w , if possible, where $u = (1, 2)$, $w = (1, -1)$

a. $v = (0, 1)$ b. $v = (2, 3)$ c. $v = (1, 4)$ d. $v = (-5, 1)$

3. Determine whether the set S is linearly independent or linearly dependent

a. $S = \{(-1, 2), (3, 1), (2, 1)\}$ b. $S = \{(-1, 2, 3), (1, 3, 5)\}$

c. $S = \{(1, -2, 2), (2, 3, 5), (3, 1, 7)\}$ d. $S = \{(-1, 2, 1), (2, 4, 0), (3, 1, 1)\}$

e. $S = \{(1, -2, 2, 1), (1, 2, 3, 5), (-1, 3, 1, 7)\}$

4. For which values of k is each set linearly independent?

a. $S = \{(-1, 2, 1), (k, 4, 0), (3, 1, 1)\}$ b. $S = \{(-1, k, 1), (1, 1, 0), (2, -1, 1)\}$

c. $S = \{(k, 1, 1), (1, k, 1), (1, 1, k)\}$ d. $S = \{(1, 2, 1, 0), (-2, 1, 1, -1), (-1, 3, 2, k)\}$

5. Find all values of m such that the set S is a basis of $\mathbb{R}^3$

a. $S = \{(1, 2, 1), (m, 1, 0), (-2, 1, 1)\}$ b. $S = \{(-1, m, 1), (1, 1, 0), (m, -1, -1)\}$

6. Find a basis for and the dimension of the subspace U

a. $U = \{(2s - t, s, s + t) \mid s, t \in \mathbb{R}\}$ b. $U = \{(s - t, s, t, s + t) \mid s, t \in \mathbb{R}\}$

c. $U = \{(0, t, -t) \mid t \in \mathbb{R}\}$ d. $U = \{(x, y, z) \mid x + y + z = 0\}$

e. $U = \{(x, y, z) \mid x + y + z = 0, x - y = 0\}$ f. $U = \text{span}\{(1, 2, 3), (2, 3, 4), (3, 5, 7)\}$

g. $U = \text{span}\{(1, 2, 4), (-1, 3, 4), (2, 3, 1)\}$ h. $U = \text{span}\{(1, 2, 1, 1), (2, 1, -1, 0), (3, 3, 0, 1)\}$

7. Find a basis for and the dimension of the solution space of the homogeneous system of linear equations.

$$\begin{array}{lll} \text{a. } \begin{cases} -x + y + z = 0 \\ 3x - y = 0 \\ 2x - 4y - 5z = 0 \end{cases} & \text{b. } \begin{cases} x + 2y - 4z = 0 \\ -3x - 6y + 12z = 0 \end{cases} & \text{c. } \begin{cases} x + y + z + t = 0 \\ 2x + 3y + z = 0 \\ 3x + 4y + 2z + t = 0 \end{cases} \end{array}$$

8. Find all values of m for which x lies in the subspace spanned by S

a. $x = (-3, 2, m)$ and $S = \{(-1, -1, 1), (2, -3, -4)\}$

b. $x = (4, 5, m)$ and $S = \{(1, -1, 1), (2, -3, 4)\}$

c. $x = (m+1, 5, m)$ and $S = \{(1, 1, 1), (2, 3, 1), (3, 4, 2)\}$

d. $x = (3, 5, 7, m)$ and $S = \{(1, 1, 1, -1), (1, 2, 3, 1), (2, 3, 4, 0)\}$

9. Find the dimension of the subspace

$$U = \text{span}\{(-2, 0, 3), (1, 2, -1), (-2, 8, 5), (-1, 2, 2)\}$$

10. Let $A = \begin{pmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 2 & 2 & 1 & 2 \end{pmatrix}$. Find $\dim(\text{col } A)$ and $\dim(\text{row } A)$

11. Which of the following are subspaces of $\mathbb{R}^3$?

(i) $\{(2+a, b-a, b) \mid a, b \in \mathbb{R}\}$

(ii) $\{(a+b, a, b) \mid a, b \in \mathbb{R}\}$

(iii) $\{(2a+b, 0, ab) \mid a, b \in \mathbb{R}\}$

12. Let $u = (1, -3, -2)$, $v = (-1, 1, 0)$ and $w = (1, 2, -3)$. Compute $\|u - v + w\|$

13. Let $u, v \in \mathbb{R}^3$ such that $\|u\| = 3$, $\|v\| = 4$ and $u \cdot v = -2$. Find

a. $\|u + v\|$ b. $\|2u + 3v\|$ c. $\|2u - v\|$

Multiple Choice

Chapter 1

1. Let $h(x) = f(g(x))$. If $f(x) = 2x - 4$ and $g(x) = x - 5$ then $h(x)$ is

- A. $2x - 14$ B. $-2x + 3$ C. $-2x - 10$ D. $-2x + 7$

2. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 5$. Find the limit (if any) $\lim_{x \rightarrow 0} \frac{f(x)}{x}$

- A. -5 B. 5 C. 0 D. None of the others

3. Find the range of the function $h(x) = (16 - x^2)^{1/2}$

- A. $[0, 4]$ B. $(0, 4)$ C. $[-16, 16]$ D. $[-4, 4]$

4. Find the domain of the function $h(x) = (16 - x^2)^{1/2}$

- A. $[0, 4]$ B. $(0, 4)$ C. $[-16, 16]$ D. $[-4, 4]$

5. If $f(x) = \begin{cases} \ln x & , 0 < x \leq 2 \\ x \ln x & , 2 < x \leq 4 \end{cases}$ then $\lim_{x \rightarrow 2} f(x)$ is

- A. $\ln 2$ B. $2 \ln 2$ C. $\ln 4$ D. Does not exist

6. If the function f continuous for all real numbers and $f(x) = \frac{x^2 - 4}{x + 2}$ when $x \neq -2$,

then $f(-2) =$

- A. -2 B. -4 C. 0 D. None of the others

7. If $f(x) = \frac{4}{x-1}$ and $g(x) = 2x$, then the solution set of $f(g(x)) = g(f(x))$ is

- A. $\left\{\frac{1}{3}, 2\right\}$ B. $\{3\}$ C. $\left\{\frac{1}{3}\right\}$ D. $\{2\}$

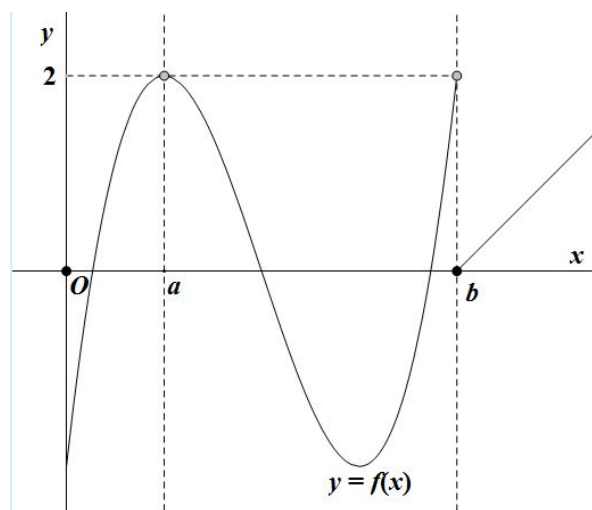
8. For $a \neq 0$, find the limit $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x^4 - a^4}$

- A. 0 B. $\frac{1}{a^2}$ C. $\frac{1}{2a^2}$ D. $\frac{1}{6a^2}$

9. If $f(x) = 2x^2 + 1$, then $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x^2} =$

- A. 0 B. 1 C. 2 D. 4

10.



The graph of the function f is shown in the figure above.

Which of the following statements about f is true?

- A. $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow b} f(x)$ B. $\lim_{x \rightarrow a} f(x) = 2$
 C. $\lim_{x \rightarrow b} f(x) = 0$ D. $\lim_{x \rightarrow a} f(x)$ does not exist

11. Let f be the function defined by the following:

$$f(x) = \begin{cases} \sin x & , x < 0 \\ x^2 & , 0 \leq x < 1 \\ 2 - x & , 1 \leq x < 2 \\ x - 3 & , 2 \leq x \end{cases}$$

For what values of x is f not continuous?

A. 2 only B. 1 only C. 0 and 2 only D. 0,1 and 2

12. Find the number k so that f is continuous at every point,

where $f(x) = \begin{cases} 8x + 3, & x < -1 \\ kx + 2, & x \geq -1 \end{cases}$

A. 7 B. -7 C. 9 D. 2

12. Find $\lim_{x \rightarrow 0} \frac{\tan x}{e^x - 1}$

A. 0 B. 1 C. -1 D. Does not exist

13. Let $f(x) = \begin{cases} \frac{x^2 - 4}{x + 2}, & x \neq -2 \\ 0, & x = -2 \end{cases}$. Which of the following statements about f are true?

(I) f has a limit at $x = -2$.

(II) f is continuous at $x = -2$

A. I only B. II only C. I and II D. None of the others

14. Determine where the function $f(x) = \sqrt{4 - |x|}$ is continuous

A. $(-\infty, -4) \cup (4, +\infty)$ B. $(-\infty, -4] \cup [4, +\infty)$

C. $(-4, 4)$ D. $[-4, 4]$

15. Let $f(x) = \begin{cases} \frac{\sqrt{1+x^2} - 1}{x^2}, & x \neq 0 \\ m, & x = 0 \end{cases}$. Find the constant m that makes f continuous on $\mathbb{R}$

A. 0 B. 1/2 C. 1/3 D. None of the others

Chapter 2

1. For $f(x) = x \sin x$, find $f'(x)$

- A. $\sin x + x \cos x$ B. $\cos x$ C. $\sin x - x \cos x$ D. $\sin x + x \sin x$

2. Find dy/dt when $x = 2$ if $dx/dt = 1$ and $y = x^3 - 3x^2$

- A. 0 B. 1 C. 2 D. 3

3. If $y = x^2 + 2$ and $u = 2x - 1$, then $dy/du =$

- A. $1/x$ B. x^2 C. x D. $6x^2 - 2x + 4$

4. What is $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\tan x}$?

- A. -1 B. 0 C. 1 D. 2

5. Find an equation of tangent to the curve $y = \frac{\sqrt{x}}{x^2 + 1}$ at $x = 1$

A. $y = -\frac{1}{4}x + \frac{3}{4}$ B. $y = \frac{1}{4}x + \frac{1}{4}$

C. $y = x - \frac{1}{2}$ D. $y = 4x - \frac{7}{2}$

6. If $x^2 + y^2 = 1$, find $\frac{dy}{dx}$

- A. $\frac{x}{y}$ B. $-\frac{x}{y}$ C. $\frac{y}{x}$ D. $-\frac{y}{x}$

7. Let f be a function such that $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = 5$.

Which of the following must be true?

(I) f is continuous at $x = 2$

(II) f is differentiable at $x = 2$

(III) The derivative of f is continuous at $x = 5$

A. I only B. II only C. III only D. (I) and (II) only

8. Let $f(x) = x\sqrt[3]{(x-2)^2}$. Find all values of c such that $f'(c) = 0$

A. 0 B. 1 C. 6/5 D. 2

9. Find the equation of the line tangent to the hyperbola $x^2 - y^2 = 16$ at the point (5, 3)

A. $y = \frac{5}{3}(x-5) + 3$ B. $y = -\frac{5}{3}(x-5) + 3$

C. $y = \frac{3}{5}(x-5) + 3$ D. $y = x - 2$

10. Find the derivative of $h(x) = xe^{2x}$

A. $2e^{2x}$ B. $2xe^{2x}$ C. $(2x+1)e^{2x}$ D. $(x+1)e^{2x}$

11. A ladder 25 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 5 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 16 ft from the wall?

A. 4/3 ft/s B. 9/4 ft/s C. 4ft/s D. 1 ft/s

12. Evaluate $\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$

A. 0 B. 1 C. $\sin x$ D. $\cos x$

13. Let f and g be differentiable functions such that $f(1) = 2$, $f'(1) = 3$, $f'(2) = -4$

$g(1) = 2$, $g'(1) = -3$, $g'(2) = 5$. If $h(x) = f(g(x))$, then $h'(1) =$

A. -9 B. -4 C. 12 D. 15

14. Let f be the function satisfying $f'(x) = x\sqrt{f(x)}$ for all real numbers x , where $f(3) = 25$. Find $f''(3)$

A. 19/2 B. 9/2 C. 5 D. 53/10

15. If $3x^2 + 2xy + y^2 = 2$, then the value of dy/dx at $x = 1$ is

- A. -2 B. 2 C. 0 D. Not defined

Chapter 3

1. Let $f(x) = 2x - 1$ for all $x \geq 2$. Select the correct one:

- A. 2 is the local minimum value B. 2 is the absolute minimum value
C. 3 is the absolute minimum value D. None of the others

2. Find two positive numbers such that the sum is 20 and the product is the largest?

- A. 10 and 12 B. 8 and 12 C. 11 and 9 D. 10 and 10

3. Find two positive numbers such that the product is 64 and the sum is the smallest?

- A. (8;8) B. (7;9) C. (16;4) D. (10; 12)

4. Find the point on the line $y = x - 4$ that is closest to the origin

- A. (1;-1) B. (2; -2) C. (3; -1) D. None of the others

5. A particle moves along the x-axis so that its velocity at time t is given by $3\sin 2t$. Assuming it starts at the origin, where is it at $t = \pi$ seconds?

- A. 0 B. $3/2$ C. $1/2$ D. $-1/2$

6. Find the points of inflection of the function $y = x^3 - 3x^2 + 1$

- A. (1;0) B. (0;1) C. (1;-1) D. ($1/2$; $3/8$)

7. Use Newton's Method with initial approximation $x_1 = 1$ to find x_3 , the third approximation to the root of the equation $x^3 + 3x - 3 = 0$. Which is the result correct to 3 decimal places?

- A. 0.818 B. 0.833 C. 0.817 D. 0.904

8. Find absolute min value of $f(x) = x^2(1-x)^3$ on $[0;2]$

- A. -4 B. 0 C. $-3/2$ D. -2

9. A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$

A. $s(t) = t^3 + 2t^2 - 6t$

B. $s(t) = t^3 + 2t^2 - 6t + 9$

C. $s(t) = t^3 + 2t^2 + 9$

D. $s(t) = s(t) = t^3 + 2t^2 + 6t + 9$

10. Find the most general anti-derivative of the function $f(x) = e^x - \frac{2}{\sqrt{x}}$

A. $F(x) = e^x - 2\sqrt{x} + C$

B. $F(x) = e^x - \frac{2}{x\sqrt{x}} + C$

C. $F(x) = e^x - 4\sqrt{x} + C$

D. $F(x) = e^x - \sqrt{x} + C$

11. Let f be the function given by $f(x) = |x|$.

Which of the following statements about f are true?

(I) f is continuous at $x = 0$

(II) f is differentiable at $x = 0$

(III) f has an absolute minimum at $x = 0$

A. I only B. II only C. III only D. I and III only

12. Find the function $f(x)$ such that $f'(x) = 3\sqrt{x}$, $f(1) = 1$.

A. $\frac{3}{2\sqrt{x}} - \frac{1}{2}$

B. $2x^{\frac{3}{2}} - 1$

C. $2x^{\frac{3}{2}} + C$

D. $\frac{1}{2\sqrt{x}} + \frac{1}{2}$

13. If $f(x) = \sin(x/2)$, then there exists a number c in the interval $\pi/2 < x < 3\pi/2$ that satisfies the conclusion of the Mean Value Theorem. Which of the following could be c ?

A. $2\pi/3$

B. $3\pi/4$

C. $5\pi/6$

D. π

14. Suppose that $f(0) = -2$ and $f'(x) \leq 3$ for all values of x . How large can $f(2)$ possibly be?

A. 1

B. 2

C. 3

D. 4

15. For the function f given by $f(x) = \frac{\ln x}{x}$ ($x > 0$). Find all the critical numbers of the function f .

A. 1

B. 0

C. e

D. e^2

Chapter 4

1. Find the average value of the function $f(x) = x^2 + 3$ on the interval $[2,5]$

- A. 48 B. 16 C. 24 D. 9.6

2. If $f(x) = \int_0^x \frac{dt}{\sqrt{t^3 + 2}}$, which of the following is false?

- A. $f(0) = 0$ B. $f'(1) = \frac{\sqrt{3}}{3}$
C. $f(-1) > 0$ D. f is continuous at x for all $x \geq 0$

3. A particle moves in a straight line with velocity $v(t) = t^2$. How far does the particle move between times $t = 1$ and $t = 2$?

- A. $1/3$ B. $7/3$ C. 3 D. 7

4. A point moves in a straight line so that its distance at time t from a fixed point of the line is $s(t) = 8t - 3t^2$. What is the total distance covered by the point between $t = 1$ and $t = 2$?

- A. 1 B. $4/3$ C. $5/3$ D. 2

5. If $\int_1^2 f(x-c)dx = 5$ where c is a constant, then $\int_{1-c}^{2-c} f(x)dx =$

- A. $5 + c$ B. 5 C. $5 - c$ D. -5

6. Given $f(x) = \begin{cases} x+1, & x < 0 \\ \cos \pi x, & x \geq 0 \end{cases}$. Calculate $\int_{-1}^1 f(x)dx$

- A. $1/2 + 1/\pi$ B. $-1/2$ C. $1/2 - 1/\pi$ D. $1/2$

7. If the position of a particle on the x -axis at time t is $-5t^2$, then the average velocity of the particle for $0 \leq t \leq 3$ is

- A. -45 B. -30 C. -15 D. -5

8. Let f be a continuous function on the closed interval $[0,2]$. If $2 \leq f(x) \leq 4$, then the greatest possible value of $\int_0^2 f(x) dx$ is

- A. 2 B. 4 C. 8 D. 16

9. Find $\frac{d}{dx} \int_2^x \sqrt{1+t^2} dt$

- A. $\sqrt{1+x^2} - 5$ B. $\sqrt{1+x^2}$
C. $\frac{x}{\sqrt{1+x^2}}$ D. None of the others

10. Use a finite sum to estimate the integral $\int_0^1 x^2 dx$ by taking the sample points to be **left** endpoints and using **four** subintervals.

- A. 0.469 B. 0.219 C. 0.333 D. 0.328

11. Use a finite sum to estimate the integral $\int_0^1 x^2 dx$ by taking the sample points to be **right** endpoints and using **four** subintervals

- A. 0.469 B. 0.219 C. 0.333 D. 0.328

12. Evaluate the integral $\int \tan 2x dx$

- A. $-\ln|\cos 2x| + C$ B. $2\ln|\cos 2x| + C$
C. $\frac{1}{2}\ln|\cos 2x| + C$ D. $-\frac{1}{2}\ln|\cos 2x| + C$

13. Find $\int \sin(2x+3) dx$

- A. $-\frac{1}{2}\cos(2x+3) + C$ B. $\cos(2x+3) + C$

C. $-\cos(2x+3)+C$ D. $\frac{1}{2}\cos(2x+3)+C$

14. For $g(x) = \int_0^x e^{1-t} dt$. Find $g'(0)$

A. 0 B. e C. 1/e D. -1/e

15. Evaluate $\int_2^3 \frac{2x dx}{x^2 + 1}$

A. $\frac{\ln 2}{2}$ B. $\ln 3 - \ln 2$ C. $\ln 2$ D. $\ln 5$

Chapter 6

1. $\int_{-1}^1 \frac{3}{x^2} dx$ is

A. 0 B. 3 C. 6 D. divergent

2. Evaluate $\int x \ln x dx$

A. $\frac{1}{2}x^2(\ln x - 1) + C$ B. $\frac{1}{4}x^2(2\ln x - 1) + C$

C. $x^2(2\ln x - 1) + C$ D. $\frac{1}{4}x^2(2\ln x + 1) + C$

3. Evaluate $\int x e^x dx$

A. $(x+1)e^x + C$ B. $x + e^x + C$

C. $x e^x + C$ D. $(x-1)e^x + C$

4. Calculate $\int_0^{\frac{\pi}{2}} e^x \sin x dx$

A. $\frac{1}{2} + \frac{e^{\frac{\pi}{2}}}{2}$ B. $-\frac{1}{2} + \frac{e^{\frac{\pi}{2}}}{2}$ C. $1 + e^{\frac{\pi}{2}}$ D. $\frac{1}{2} - \frac{e^{\frac{\pi}{2}}}{2}$

5. Find $\int \ln x dx$

A. $x \ln x - x + C$ B. $-x \ln x + x + C$
 C. $x \ln x + x + C$ D. $\frac{1}{x} + C$

6. $\int x \cos x dx =$

A. $x \sin x - \cos x + C$ B. $x \sin x + C$
 C. $x \sin x + \cos x + C$ D. $-x \sin x + \cos x + C$

7. Estimate the area under the graph of $f(x) = x^2 + 1/x$ from $x = 1$ to $x = 3$ using 4 rectangles of width 0.5 and the midpoint rule for approximating the area.

A. 9.715 B. 11.7 C. 9.765 D. 8.033

8.

x	0	0.5	1	1.5	2
$f(x)$	3	3	5	8	13

A table of values for a continuous function f is shown above. If four equal subintervals of $[0,2]$ are used, which of the following is the trapezoidal approximation of $\int_0^2 f(x) dx$?

A. 8 B. 32 C. 16 D. 12

9. $\int_1^{\infty} \frac{x dx}{(1+x^2)^2}$ is

A. -1/2 B. 1/2 C. 1/4 D. divergent

10. Evaluate $\int_0^1 \frac{dx}{\sqrt{x}}$

A. 0 B. 1 C. 2 D. 4

11. Evaluate $\int_1^{\infty} \frac{2x}{x^2+1} dx$

- A. $16\ln 11$ B. $10\ln 10$ C. 1 D. It is divergent

12. Which integrals are convergent

(I) $\int_1^{\infty} \frac{x^2 dx}{x^3+1}$ (II) $\int_2^{\infty} \frac{dx}{x^2-1}$

- A. I only B. II only
C. Both I and II D. None of the others

13. Which integrals are convergent

(I) $\int_1^{\infty} \sin t dt$ (II) $\int_2^5 \frac{dx}{x^2-9}$

- A. I only B. II only
C. Both I and II D. None of the others

14. Calculate $\int_0^{\frac{\pi}{2}} \frac{\sin x dx}{\sin x + \cos x}$

- A. 0 B. π C. $\pi/4$ D. $\pi/2$

15. Evaluate $\int_0^{\infty} \frac{x}{e^x} dx$

- A. 1 B. $1/2$ C. 0 D. It is divergent

Chap 1b

1. Find all solutions of the following system of linear equations $\begin{cases} x - y - z = 3 \\ -x - y + z = -1 \end{cases}$

A. $x = 3, y = -1, z = 1$

B. $x = 1, y = -1, z = -1$

C. $x = t - 2, y = -1, z = t$

D. $x = t, y = -1, z = t - 2$

2. Find the system of linear equations whose augmented matrix is given as

$$\left[\begin{array}{cccc} 1 & -2 & 0 & 6 \\ -3 & 1 & 5 & 2 \\ 0 & 1 & 3 & 4 \end{array} \right]$$

A. $\begin{cases} x - 2y = 6 \\ 3x - y - 5z = -2 \\ y + 3z = -4 \end{cases}$

B. $\begin{cases} x - 2y = 6 \\ 3x - y - 5z = 2 \\ y + 3z = 4 \end{cases}$

C. $\begin{cases} x - 2y = 6 \\ 3x - y - 5z = -2 \\ y + 3z = 4 \end{cases}$

D. $\begin{cases} x - 2y + 6t = 0 \\ -3x + y + 5z + 2t = 0 \\ y + 3z + 4t = 0 \end{cases}$

3. Find all values of m such that the following system has no solution

$$\begin{cases} x + y - z = 2 \\ 3x - y + 2z = 3 \\ 2x - 2y + 3z = m \end{cases}$$

A. Any number

B. All numbers but 1

C. 1

D. 7

4. Let A be the augmented matrix of a homogeneous of 3 equations in 6 variables. If $\text{rank}(A) = 1$, how many solutions and how many parameters does this system have?

A. Infinitely many solutions and 3 parameters

- B. Infinitely many solutions and 2 parameters
- C. Infinitely many solutions and 5 parameters
- D. Unique solution

5. Find all values m such that the system of equations
$$\begin{cases} x + y - z = 1 \\ x + 2y + mz = 0 \\ 2x + 3y - 2z = m \end{cases}$$
 has exactly one solution

- A. $m \neq 1$
- B. $m \neq 2$
- C. $m \neq -1$
- D. $m = -1$

Chap 2b

1. If ABC can be formed and A is 4×4 , C is 7×7 . What is the size of B ?

- A. 4×7
- B. 4×4
- C. 7×4
- D. 7×7

2. Let $T : R^2 \rightarrow R^2$ be a linear transformation such that $T(u) = (1, 2)$, $T(v) = (-1, 0)$ for given $u, v \in R^2$. Find $T(2u - 3v)$

- A. $(-2, 8)$
- B. $(-2, 4)$
- C. $(1, 0)$
- D. $(5, 4)$

3. Let $A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix}$. Solve $AXB = BA$, where X is a matrix

- A. $X = I$
- B. $X = \begin{pmatrix} 59 & 32 \\ -24 & -13 \end{pmatrix}$

- C. $X = \begin{pmatrix} 27 & -16 \\ -32 & 19 \end{pmatrix}$
- D. None of the others

Chap 3b

1. The characteristic polynomial of $A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$ is

- A. $(x - 2)(x + 1)$
- B. $x^2 - 3x + 2$
- C. $(x + 2)(x + 1)$
- D. x^2

2. Let $A = \begin{pmatrix} 1 & * & * & * \\ 0 & 3 & * & * \\ 0 & 0 & 5 & * \\ 0 & 0 & 0 & 7 \end{pmatrix}$, where $(*)$ denotes any real number. Compute $\det(2A^{-1})$

A. $\frac{2}{105}$

B. 210

C. $\frac{16}{105}$

D. None of the others

3. Give that $\lambda = 1$ is an eigenvalue for the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Find a set of basic eigenvectors corresponding to this eigenvalue $\lambda = 1$

A. $\{(0,0,1)\}$

B. $\{(1,0,0), (0,0,1)\}$

C. $\{(1,0,0)\}$

D. $\{(0,-1,1)\}$

4. Find m such that the matrix $\begin{pmatrix} 0 & m & -4 \\ 2 & 3 & -1 \\ 1 & 4 & 1 \end{pmatrix}$ is not invertible

A. All numbers but $-20/3$

B. All numbers but $20/3$

C. $20/3$

D. $-20/3$

5. Find the (1,2) - cofactor of the matrix $\begin{pmatrix} 1 & 2 & 3 \\ 4 & -1 & 5 \\ 0 & 7 & 6 \end{pmatrix}$

A. 24

B. -24

C. 9

D. $\begin{pmatrix} 4 & 5 \\ 0 & 6 \end{pmatrix}$

6. Find the first row of adjugate of the matrix $\begin{pmatrix} 0 & 2 & -4 \\ 2 & 3 & -1 \\ 1 & 4 & 1 \end{pmatrix}$

A. [7, 18, 10]

B. [7, -18, 10]

C. [7, -3, 5]

D. [7/26, 9/13, -5/13]

7. Find all the eigenvalues of the matrix $\begin{pmatrix} 2 & 1 & 5 \\ 0 & 1 & 1 \\ 0 & -8 & 7 \end{pmatrix}$

- A. 2,1,7 B. 2,3,5 C. 2,2,6 D. None of the others

Chap 5b

1. Let A be a 3×5 matrix. Choose correct statements

- (i) A can have rank 3
 (ii) A can have rank 5
 (iii) A can have linearly independent rows
 (iv) A can have linearly independent columns
- A. (i) only B. (i) and (iii) only
 C. (ii) and (iv) only D. (iv) only

2. Let $U = \{(a, b, c, d) \mid 3a - 5b = 0, b + c + d = 0\}$ be a subspace of $\mathbb{R}^4$.

Find the dimension of U

- A. 1 B. 2 C. 3 D. 4

3. Find x such that the set $\left\{ \begin{pmatrix} 2 \\ x \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$ is independent

- A. $x \neq -1$ B. $x = 1$ C. $x \neq 2$ D. None of the others

4. Let $U = \{(x, y, z) \mid 2x - y + z = 0\}$ be a subspace of $\mathbb{R}^3$. Which of the following statements are true?

- (i) $U = \text{span}\{(1, 0, -2), (0, 1, 1)\}$
 (ii) $U = \text{span}\{(1, 2, 0)\}$

- A. (i) only B. Both (i) and (ii) C. (ii) only D. None

5. Which condition on the numbers a, b, c is the vector $(a, b, c) \in \text{span}\{(1, 0, 2), (1, 2, 8)\}$

- A. $c = 2a + 3b$ B. $c = -2a - 3b$ C. $c = 2a - 3b$ D. $c = -2a + 3b$

6. Find all values of m such that the set $\{(2, m, 1), (m, 0, 0), (1, 1, m)\}$ is a basis of $\mathbb{R}^3$

- A. $m \neq 0$ B. $m \neq \pm 1$ C. $m \neq 1$ D. $m \in \mathbb{R} \setminus \{0, 1, -1\}$

Test

1. Find the inverse of the matrix $\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$

- a. $\begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}$ b. $\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$ c. $\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$ d. $\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$

2. Find the transpose of the matrix $\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$

- a. $\begin{pmatrix} -1 & 1 \\ 3 & -2 \end{pmatrix}$ b. $\begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$ c. $\begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}$ d. -1

3. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ and $f(x) = x^2 - 3x + 2$. Find $f(A)$

- a. $\begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}$ b. $-6I$ c. $6I$ D. None of the others

4. For the function $f(x) = \frac{2x+3}{x-1}$. Evaluate the limit $\lim_{x \rightarrow 3} f(x)$

- A. 6 B. 2 C. 3 D. 9/2

5. Let $f(x) = 2x + 3$ and $g(x) = \sqrt{x}$. Find $g \circ f(x)$

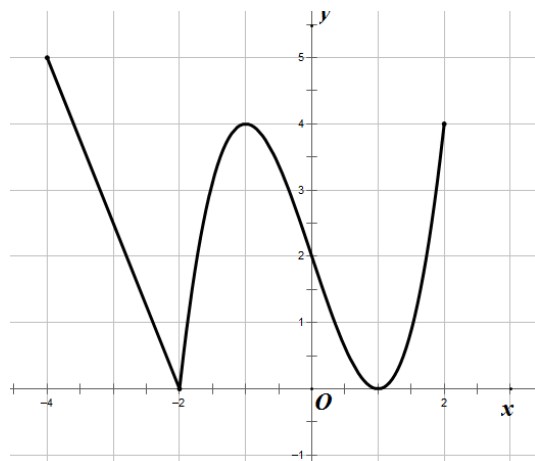
- A. $\sqrt{2x+3}$ B. $2\sqrt{x} + 3$
C. $\frac{\sqrt{x}}{2x+3}$ D. $(2x+3)\sqrt{x}$

6. For $y = x^3 + x^2 - 2$, find dy/dx

A. 1 B. $3x^2 + 2x$

C. $\frac{x^4}{4} + \frac{x^3}{3} - 2x + C$ D. 5

7. The graph of the function f is given. Find the absolute maximum



A. 5 B. -4 C. 4 D. 0

8. Evaluate $\int_0^1 xe^{-x} dx$

A. $1 - \frac{2}{e}$ B. $1 + \frac{2}{e}$ C. $\frac{1}{2}$ D. e^{-1}

9. Evaluate the improper integral if it exists

$$\int_1^{\infty} \frac{1}{x^3} dx$$

A. 1 B. 1/4 C. 1/2 D. It diverges

10. Solve for z in the system of equations

$$\begin{cases} 2x - y + z = 3 \\ x + y - z = 0 \\ -x + y + 2z = 7 \end{cases}$$

A. 1 B. 2 C. 3 D. 6

11. Which of the following matrices are in reduced row-echelon form?

$$(i) \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$(ii) \quad N = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

- A. (i) only B. (ii) only C. Both (i) and (ii) D. Either (i) nor (ii)

12. If $\begin{vmatrix} 1 & 4 \\ 3 & k \end{vmatrix} = 6$, then $k = ?$

- A. -6 B. 6 C. 18 D. 24

13. Let A be a 3×3 matrix such that $\det(A) = -2$. Find $\det(2A^T)$

- A. -4 B. -2 C. -6 D. -16

14. Let $u = (1, 1, 3)$ and $v = (-1, 2, 1)$ in $\mathbb{R}^3$. Compute $\|u + v\|$

- A. 5 B. 25 C. $(0, 3, 4)$ D. None of the others

15. Let $u = (1, 2)$, $v = (3, 4)$ and $x = (2, 2)$ in $\mathbb{R}^2$. Find (a, b) such that $x = au + bv$.

- A. $(1, 1)$ B. $(-1, 1)$ C. $(1, -1)$ D. $(-1, -1)$

16. Let f be a function that is continuous on $[2, 4]$ with $f(2) = 10$ and $f(4) = 20$.

Which of the following statements is true?

A. $f(x) = 13$ has at least one solution in $(2, 4)$.

B. $f(3) = 15$.

C. f attains a maximum on $(2, 4)$.

D. $f'(x) = 5$ has at least one solution in $(2, 4)$.

E. $f'(x) > 0, \forall x \in (2, 4)$

17. If $\int_{-1}^0 \frac{18dx}{x^3 - 3x + 2} = a + b \ln 2$ ($a, b \in \mathbb{Z}$), then $a + b = ?$

- A. 1 B. -2 C. -5 D. 4 E. 7

18. Let $f(x) = \begin{cases} \frac{e^{2mx} - 1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$. If f is differentiable at zero, then $f'(0) = ?$

- A. 1 B. -1 C. 1/2 D. 2 E. Does not exist

19. Let M and N be $n \times n$ matrices such that $MN = 0$.

Which of the following statements is true?

- A. $M^2 - N^2 = (M + N)(M - N)$ B. $N^2 M^2 = 0$ C. $NM = 0$
D. $(NM)^2 = 0$ E. All of these are true

20. One of the eigenvalues of the diagonalizable matrix $A = \begin{pmatrix} 3 & 104 & -339 \\ 0 & -34 & 120 \\ 0 & -10 & 36 \end{pmatrix}$ is 3.

Find the sum of other two eigenvalues of A .

- A. 2 B. 1 C. 0 D. -1 E. 10

21. Let A be an $n \times n$ matrix. Which one of the following statements is not equivalent to the other four?

- (i) A is not invertible.
(ii) The equation $AX = b$ has a unique solution X for any n -vector b :
(iii) The rows of A are linearly independent.
(iv) A can be row-reduced to the identity matrix I_n .
(v) The column rank of A is n .
A. (i) B. (ii) C. (iii) D. (iv) E. (v)